

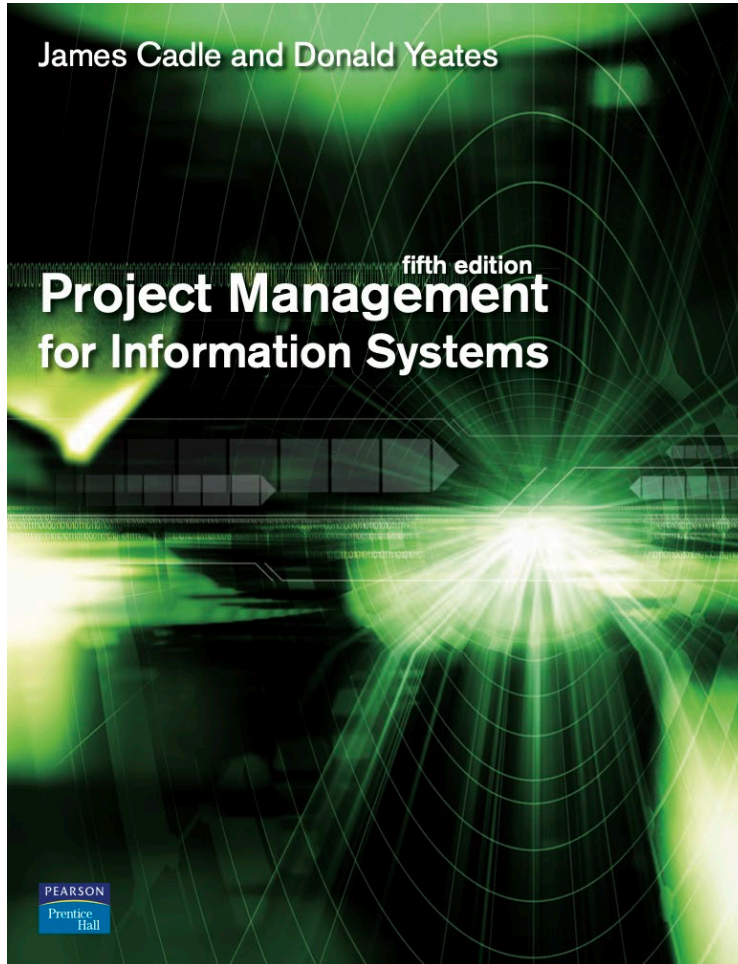
BU5300 – IT Project Management

Lecture 7: Monitoring and Controlling IT Projects & Reporting Progress

Chapters 11 - 13

Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11, 12 and 13

Lecture 8 - Chapters 14 and 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

Learning Outcomes

- List the three perspectives against which projects should be managed
- Define the structure for a project timesheet
- Describe six kinds of quality control review
- Explain how earned value analysis can be used to monitor progress

Chapter 11: Monitoring Progress

Project Lifecycle



Monitoring Progress - Overview

- Measuring project progression and performance, and ensuring that everything happening aligns with the project management plan
- The project needs to be managed from three perspectives, those of the triple constraint of **time**, **cost** and **quality**
- In this chapter, we consider monitoring techniques for each element of the triple constraint

Monitoring effort

- The major resource element is staff time
 - Typically collected by asking the team to fill out time sheets
 - Time sheets may be the basis of billing the customer
 - For internal project, time sheets are used to track resource use
 - We get a much more accurate record if people fill in their timesheets daily

$(\text{Original estimate}) - (\text{Effort expended to date}) = \text{Estimate to go}$

- Other costs to be considered for resources are machine usage and bought-in hardware and services

Effort-monitoring timesheet

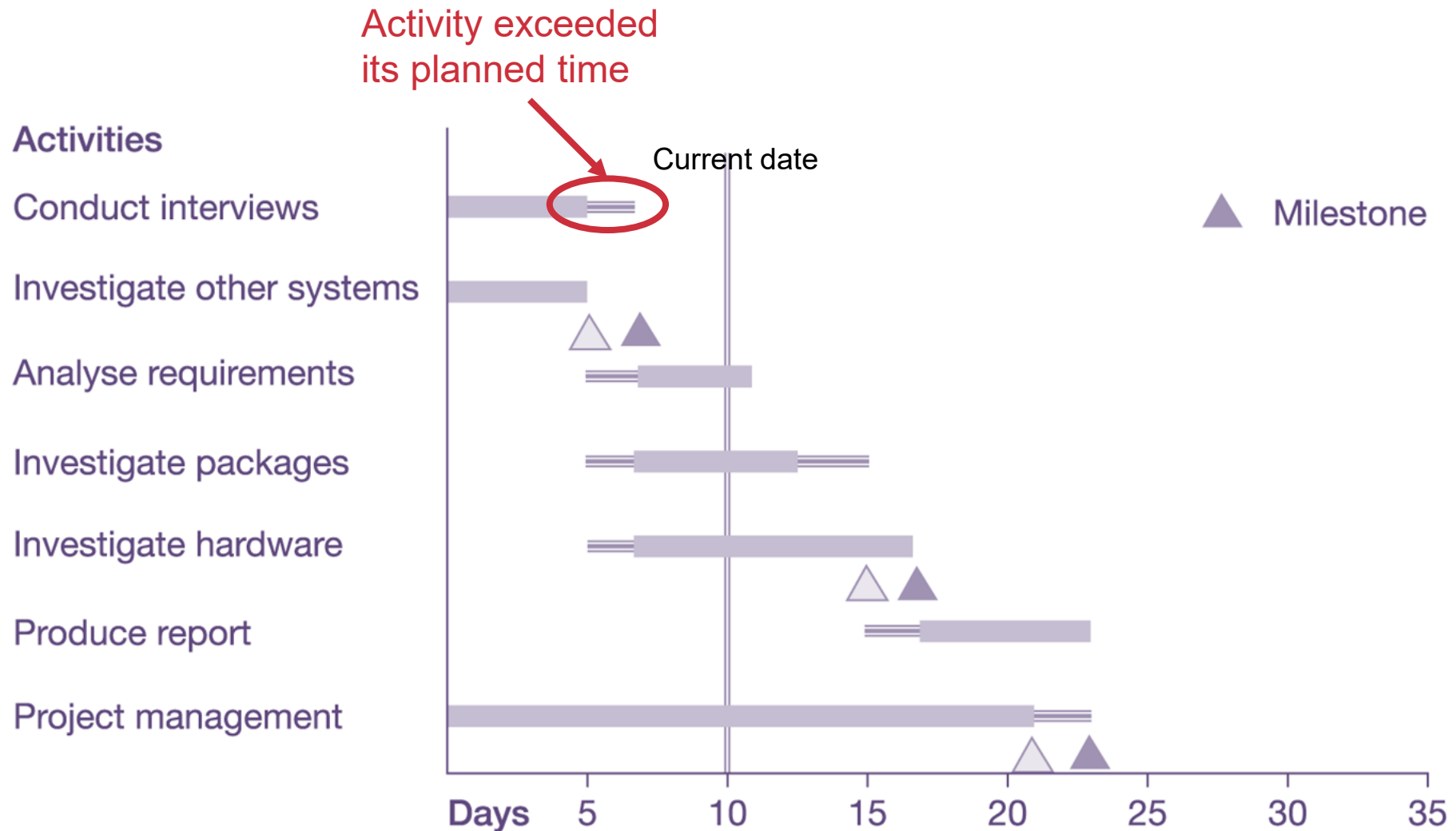
Name: <i>DAVE SIMS</i>		Project: <i>PERSONNEL</i>		Week ending: <i>19 May</i>						
Code		Mon	Tue	Wed	Thu	Fri	Sat	Sun	Total	To go
<i>A/01</i>	<i>CODE PROGRAM CV004</i>	<i>6.5</i>	<i>5.0</i>	<i>7.5</i>					<i>19.0</i>	<i>NIL</i>
<i>A/02</i>	<i>TEST PROGRAM CV004</i>				<i>7.5</i>	<i>3.0</i>			<i>10.5</i>	<i>NIL</i>
<i>A/07</i>	<i>CODE PROGRAM EN025</i>					<i>4.0</i>			<i>4.0</i>	<i>15.0</i>
<i>M/03</i>	<i>TEAM MEETING</i>	<i>1.0</i>							<i>1.0</i>	
<i>M/02</i>	<i>COMPLETE TIMESHEET</i>					<i>0.5</i>			<i>0.5</i>	

Project-Level Effort Monitoring

- Having collected information at an individual level, the next step is to summarize it at a project level to see the overall situation
- Project planning packages usually have facilities to do this
- For each activity, we need to know:

Not Started (planned)	In Progress (current or estimated)	Completed
Planned start and end date	Start date <i>Did we start in time?</i>	End date <i>Did we delivered in time?</i>
Effort and cost estimates	Effort and cost (done + still to go) <i>Are we on the right track?</i>	Total effort and cost done <i>Are our estimates precise?</i>

Bar chart illustrating progress



Possible expenses to consider

- Project-specific training
 - for example, in the programming language or environment
- Travel
- Specially arranged accommodation
- Lodging and subsistence costs
- Consumables
- Insurance

Monitoring quality

- When to do quality control?
 - at the project milestones
 - when moving from one development phase to the next
 - when found a defect or fault
- But also...
 - first application of new standards
 - applying new techniques or methods
 - recruiting inexperienced or new staff
- Reviewers must be clear about the criteria to be applied (pass or fail)
- Methods for Monitoring quality
 - Self-checking by author
 - Peer review
 - Team leader review
 - Walkthrough
 - Fagan inspection
 - External review

Monitoring Quality

Quality checklists

Specific checklists depend on the nature of the project and the requirements of the stakeholders

- Code Quality checklist
- Testing checklist
- Documentation checklist
- Security checklist
- Performance checklist
- Accessibility checklist

PROJECT QUALITY CHECKLIST TEMPLATE

DEFECT ID <i>A unique ID so that the bug can be referenced</i>		AUTHOR <i>The person writing the bug report</i>		RELEASE BUILD NO. <i>The release build number of the software in which the bug was detected</i>	
OPEN DATE		PROBLEM AREA		PROBLEM DESCRIPTION	
CLOSE DATE		PROBLEM TITLE		CURRENT ENVIRONMENT	
DEFECT TYPE	X	WHO DETECTED	X	HOW DETECTED	X
Functionality		Quality Assurance		Testing	
Architectural		External Customer		Review	
Connectivity		Internal Customer		Walkthrough	
Consistency		Development		JAD	
Database Integrity		PRIORITY	X	STATUS	X
Documentation		Critical		Open	
GUI		High		Being Reviewed by Development	
Installation		Medium		Returned by Development	
Memory		Low		Ready for Testing in the Next Build	
Performance		SEVERITY	X	Closed (QA)	
Security and Controls		Critical		Returned by (QA)	
Standards and Conventions		High		Deferred to the Next Release	
Stress		Medium		ASSIGNED TO	
Usability		Low			
STATUS DESCRIPTION		FIXED BY		PLANNED FIX BUILD NO.	

Monitoring Quality

Agile's Definition of Done (DoD)

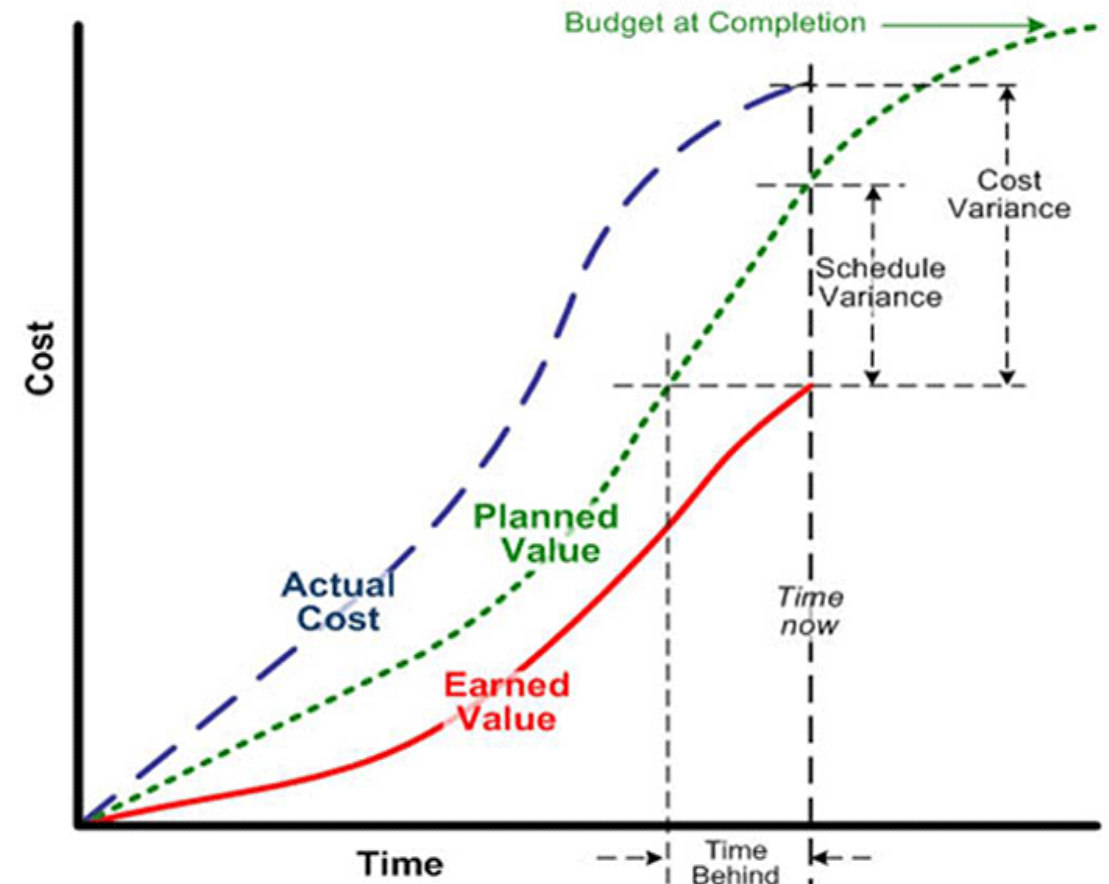
- Criteria that a **particular task or user story** must be met to be considered complete
- Typically includes a set of objective standards and guidelines
 - The team should create a checklist that must be passed before delivery
- Helps ensure that work is completed to a consistent standard of quality and completeness

DoD Checklist example

- ☐ Product owners accepts the User Story
- ☐ Passed unit testing
- ☐ Code review
- ☐ Meet acceptance criteria
- ☐ Passed functional testing
- ☐ Meet non-functional requirements

Earned value analysis

- **Planned Value (PV):** the budget that has been approved for the work scheduled, aka BCWS
- **Earned Value (EV):** the actual work that has been completed by the specified date, aka BCWP
- **Actual Cost (AC):** the actual total costs incurred for the work completed within the specified date, aka ACWP
- **Cost variance (CV):** $AC - EV$
- **Schedule variance (SC):** $PV - EV$
- **Schedule Performance Index (SPI):** EV / PV
- **ScCost Performance Index (SPI):** AC / PV
- **Estimate Time to Complete (ETC):**
original time planed / SPI

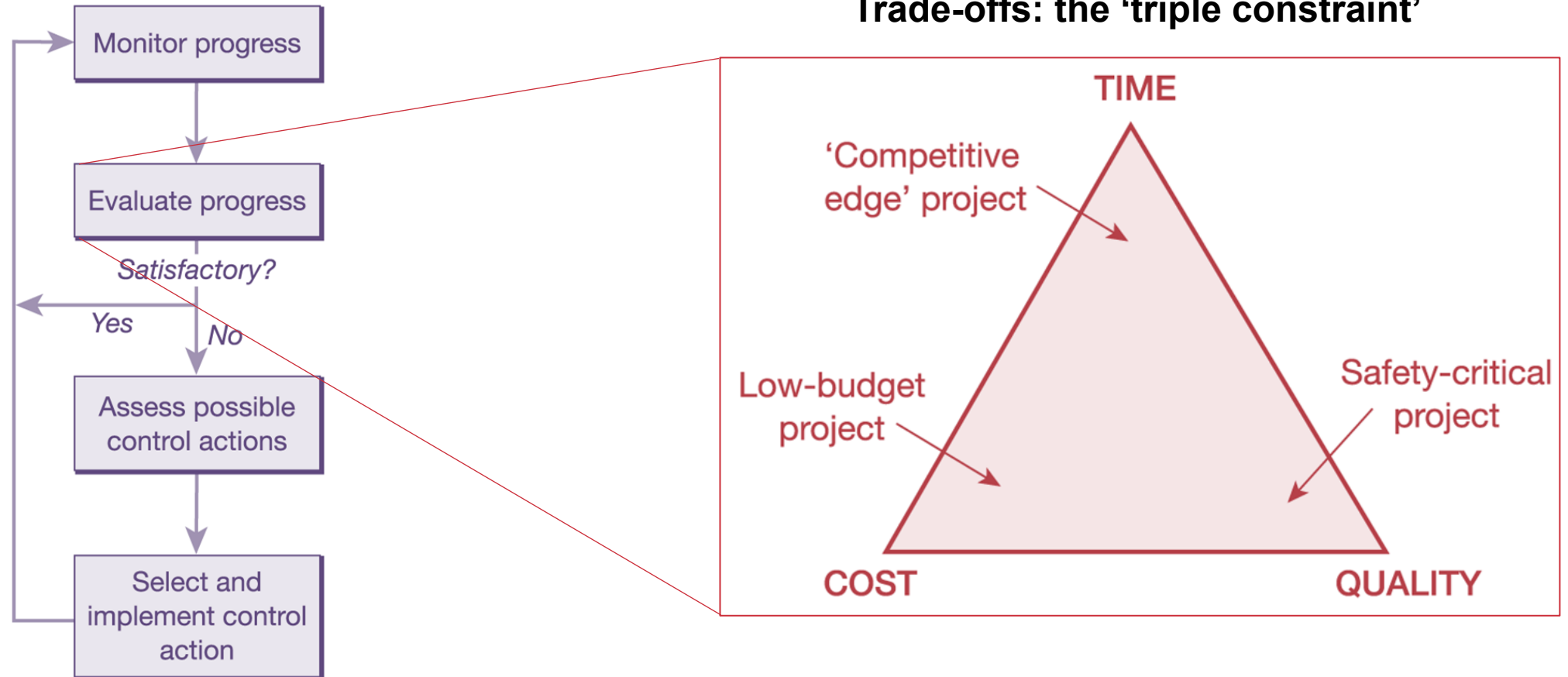


Chapter 12: Exercising Control

Learning Outcomes

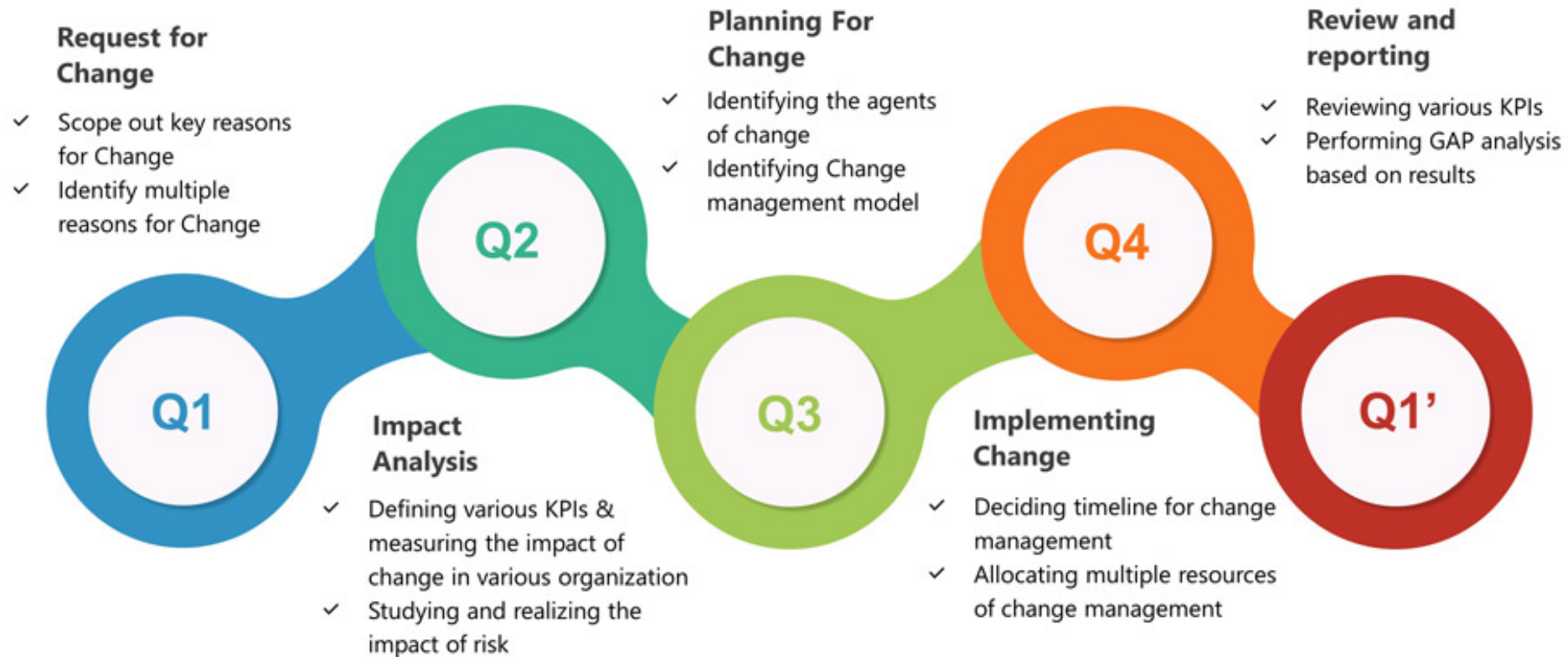
- Describe a generic model for controlling project work
- List different kinds of corrective action to get a project 'back on course'
- Distinguish between change control and configuration management

Monitoring and control cycle



Change control

How to implement and manage changes?



Possible corrective actions

- Add more staff
- Add different skills
- Use overtime
- Reassign tasks
- Increase supervision
- Decrease supervision
- Find improved methods
- Streamline procedures
- Change resource priorities
- Re-plan the project
- Change the phasing of deliverables
- Decrease inspections
- Increase inspections
- Encourage the team
- Introduce incentives
- Subcontract some work
- Negotiate changes to the specification

Change control and configuration management

How is that different?

Change control

- Set of procedures to ensure that changes made only after a consideration of the full impacts
- Organizational tools to help individuals make successful transition
- Project perspective

Configuration management

- Version control to ensure that all changes are properly recorded and managed
 - and so that knock-on effects on other projects can be identified
- Identify and monitor changes on the functional characteristics
- Product perspective

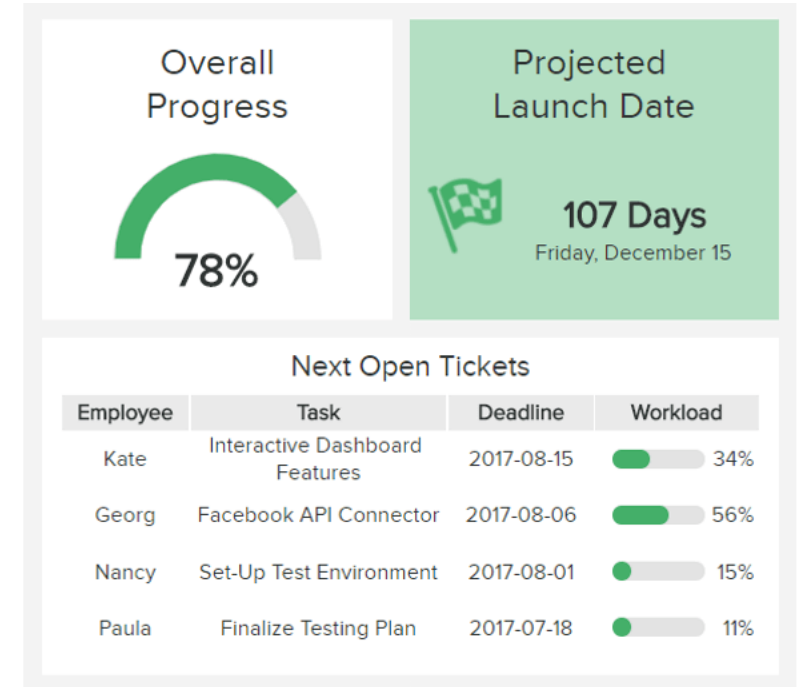
Chapter 13: Reporting Progress

A few questions worth discussing

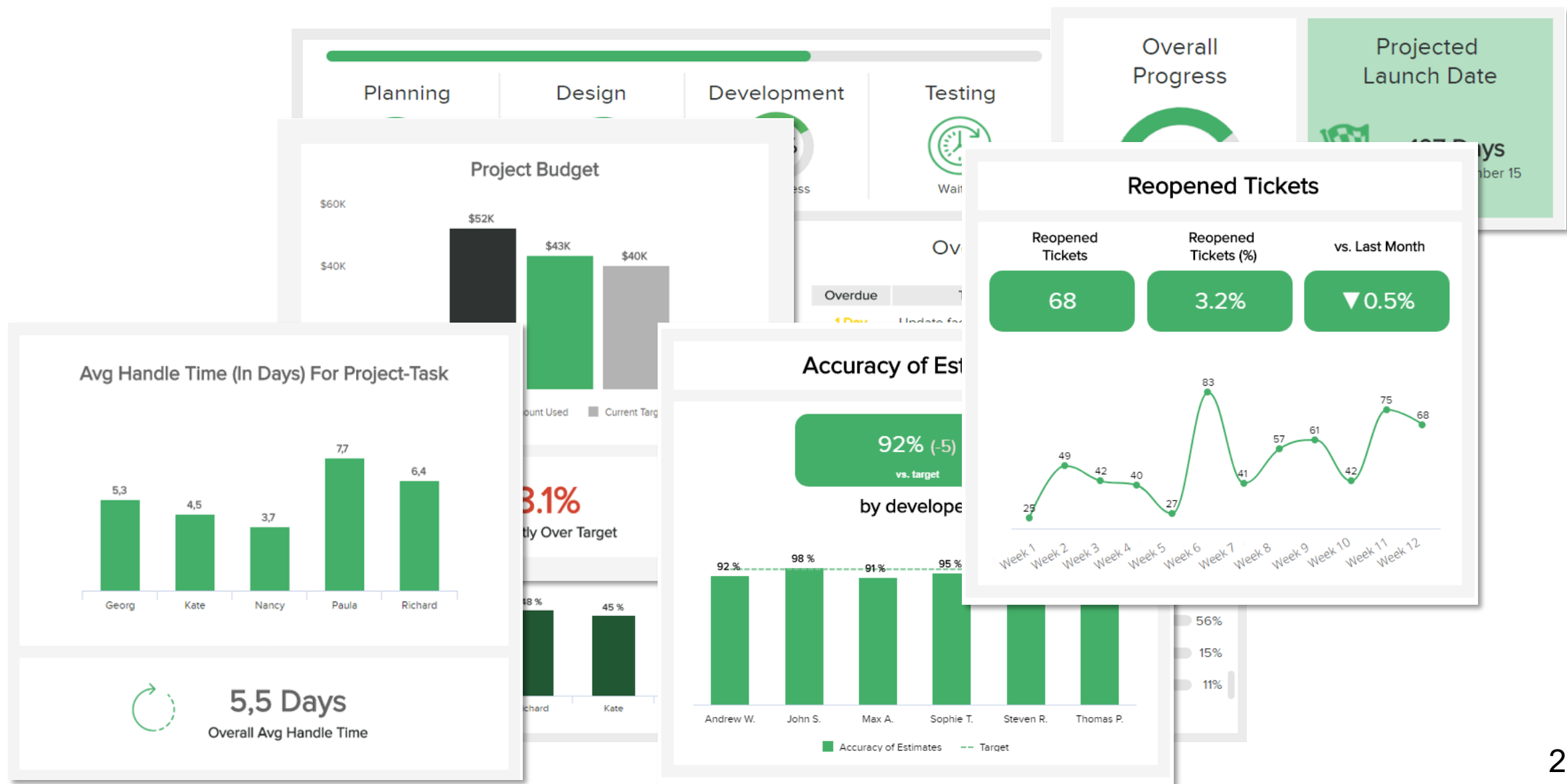
- Who is the report directed to?
 - The contents of the report and level of detail should satisfy the needs of the recipients
- How about frequency of reporting?
 - Monthly; any other week (time-oriented)
 - End of a phase or sprint (process-oriented)
 - Live reports
- How to present the report?
 - Written report
 - Face-to-face presentation
 - Status dashboard (autonomous reporting)

Report contents (typical)

- Period covered
- Narrative summary of progress
- Milestones achieved/deliverables completed
- Problems encountered (and solutions)
- Projected completion date
- Costs to date and predicted
- Changes identified and implemented



Progress Status Dashboard



Progress Report

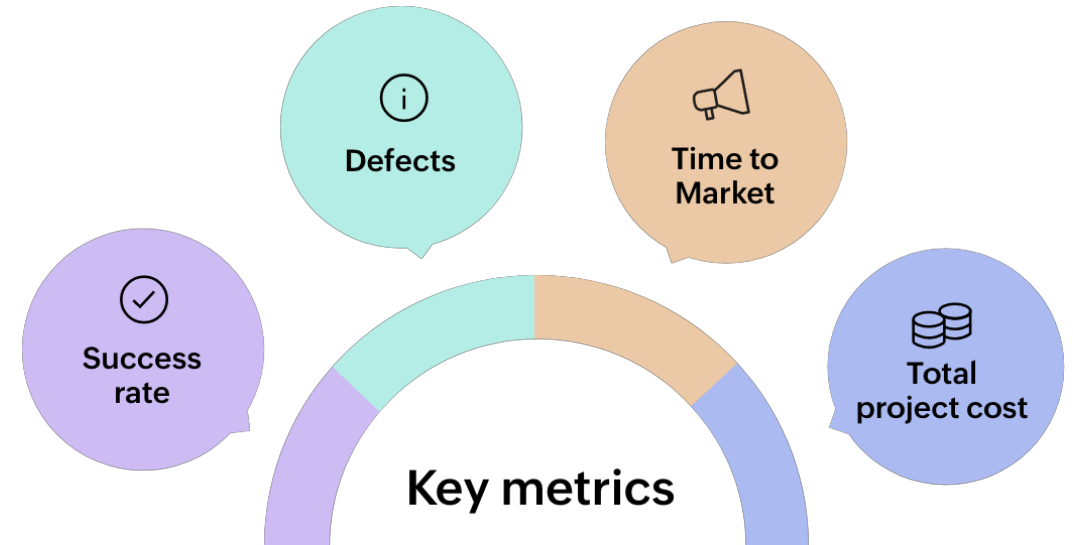
PRINCE 2 vs PMI

Document:	Highlight report # 4	Project:	Pen Project	Author:	Project Manager	Date:
1. Date and period of the report						
End of the second week of Stage 2, for a period of one week.						
2. Status summary						
Time: +5% Cost: ±0 The Project is going well so far.						
3. Work Packages in this period						
- Selected supplier: Completed - Catalogs of pens: Completed (behind schedule) - Pens short list to be evaluated: Completed (behind schedule) - Evaluation pens purchase order: Not finished (behind schedule) - Sample evaluation pens arrived: Not started (behind schedule) - Logo: Completed It took three extra days for the supplier to send us the catalogs.						
4. Work Packages for the next period						
- Purchase order of pens to be evaluated (moved from this period) - Sample pens for evaluation (moved from this period) - List of evaluators ✖ - Evaluation form ✖ - Next stage plan ✖ We are going to get help from IT department to conduct these two packages faster and catch up on the planned schedule.						
5. Stage tolerances						
Time: +5%, this is inside the tolerances set for the Stage						
Cost: ±0%						
		No exception yet.				
6. Change requests						
We fell behind schedule because the supplier sent us catalogs three days later than we expected. We designed a corrective action to get help from the IT department for two of our future Work Packages. We will recover time by this. (approved by Project Manager)						
7. Key issues and risks						
We might have more delays in receiving evaluation pens and the final pens from the supplier.						

[illegible]

Reporting Progress in Agile Project

- Reports provide agile teams with greater visibility
 - Display team efforts
 - Monitor the progress toward key goals, and
 - Allows effective planning
- For **teams**, it shows how to incorporate timely changes so they can keep up with changing business environments
- For **product owners**, it helps to initiate constructive conversations, and set standards with team and external stakeholders

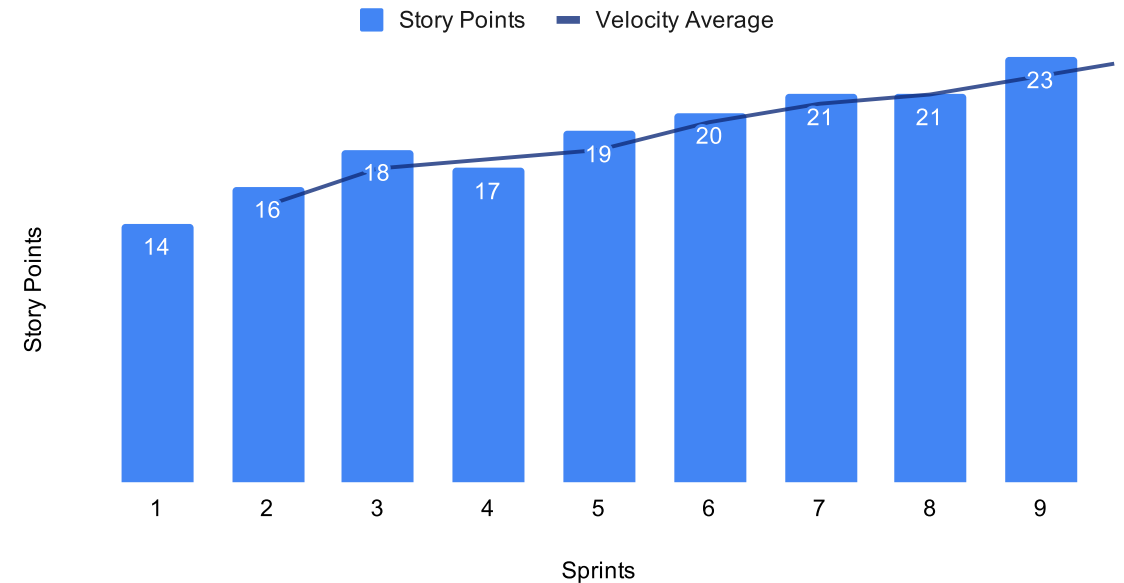


Velocity

Effectiveness of the Development Process

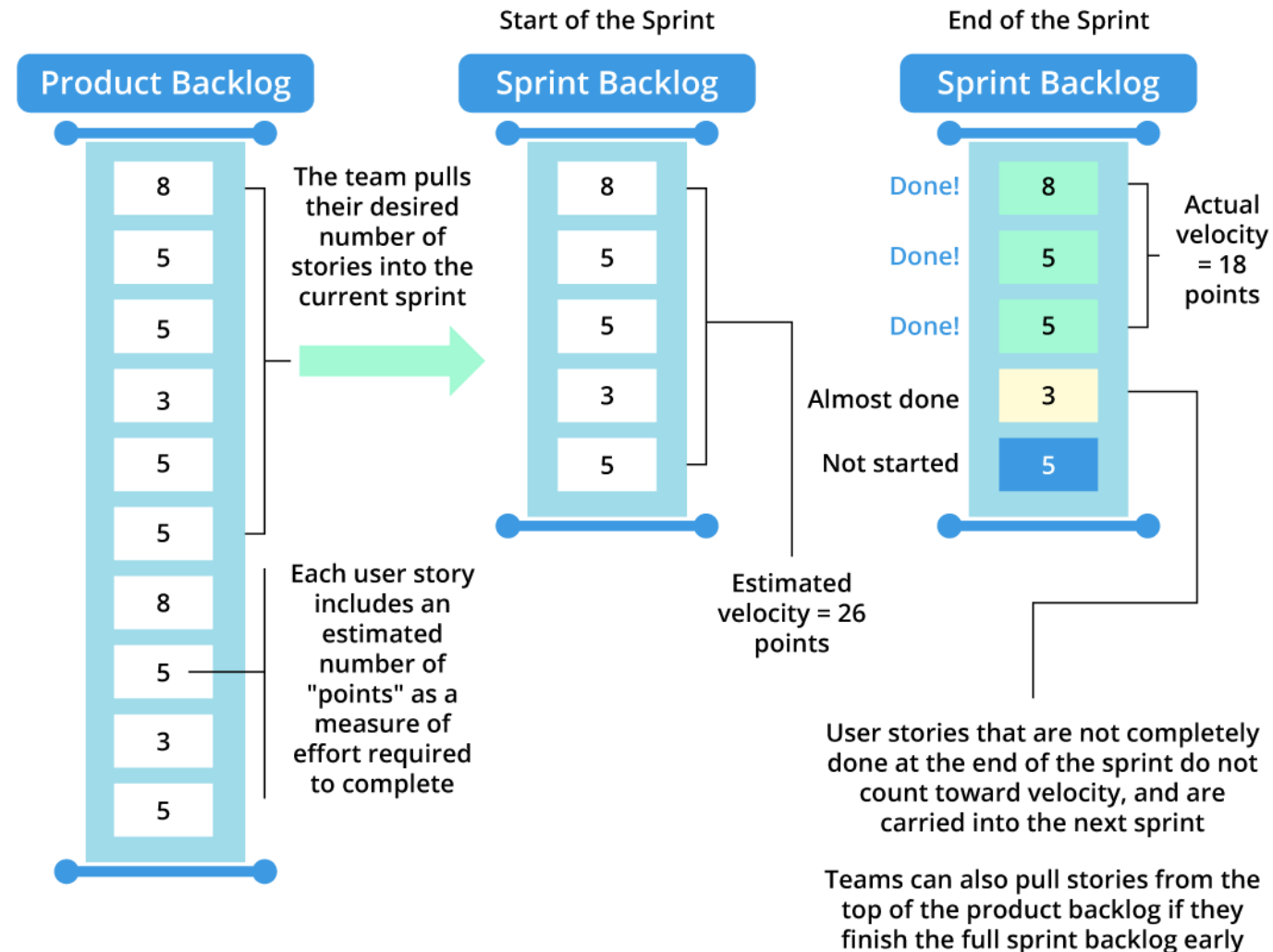
- **Velocity** is the key metric in SCRUM I
- It measures the average story points that a team delivered over the past few sprint cycles
- With a velocity report, **you can calculate the amount of work your team can handle in the upcoming sprints**

Velocity



Velocity

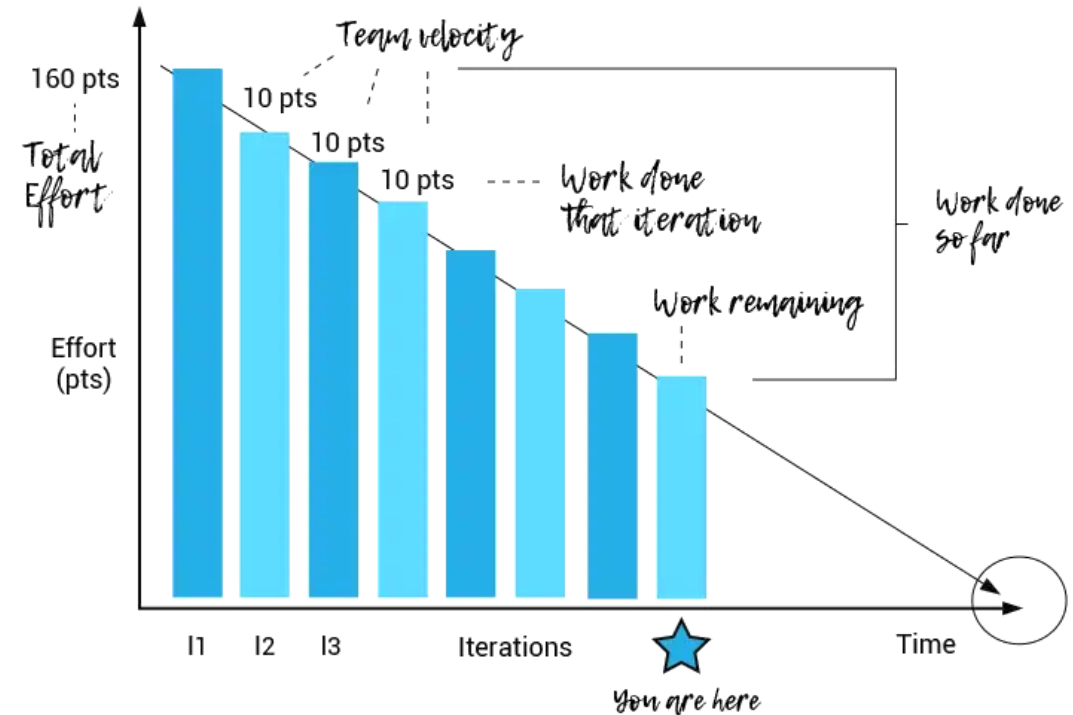
Effectiveness of the Development Process



Sprint Burndown

Effectiveness of the Development Process

- **Sprint Burndown** is the best way to monitor project accomplishment
- It provides a visual and almost real-time update of **how many story points your team has completed throughout the sprint**



Epic and Release Burndown

Effectiveness of the Development Process

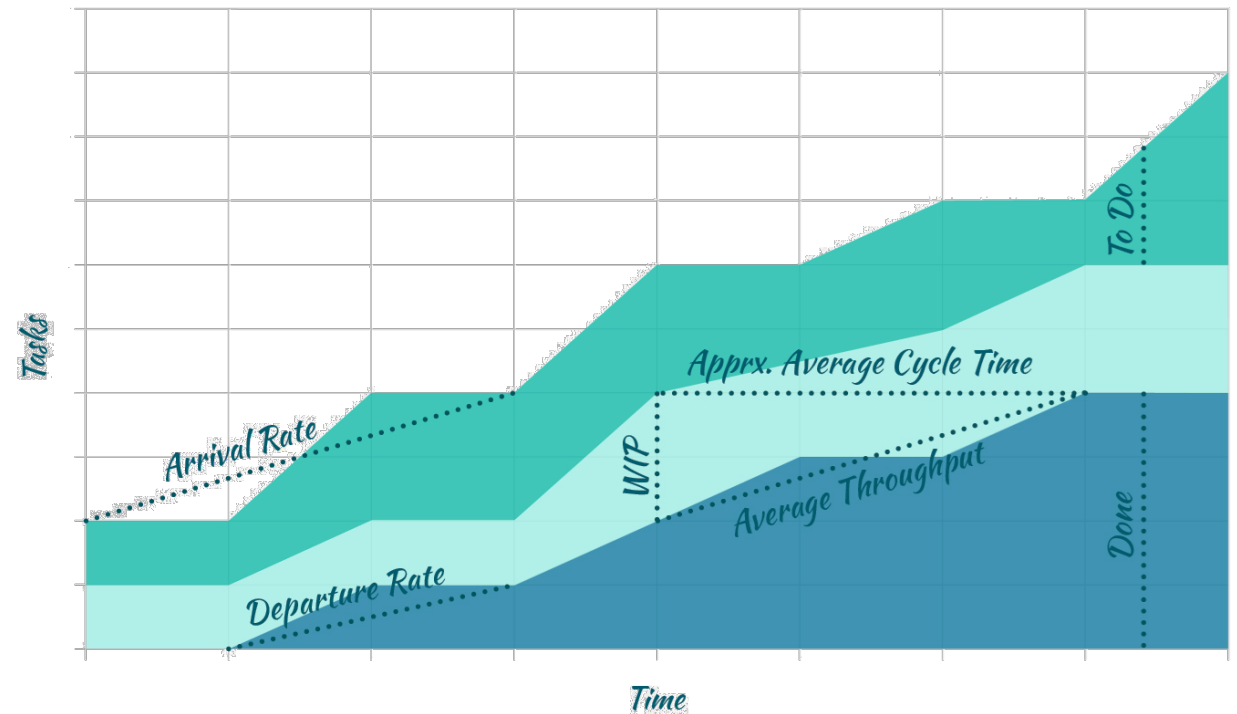
- **Epic and Release Burndown** offers a big picture view of your project status as sprints usually address tasks from different epics
- Most importantly, this big picture project reporting helps to **keep scope under control**



Cumulative Flow Diagram

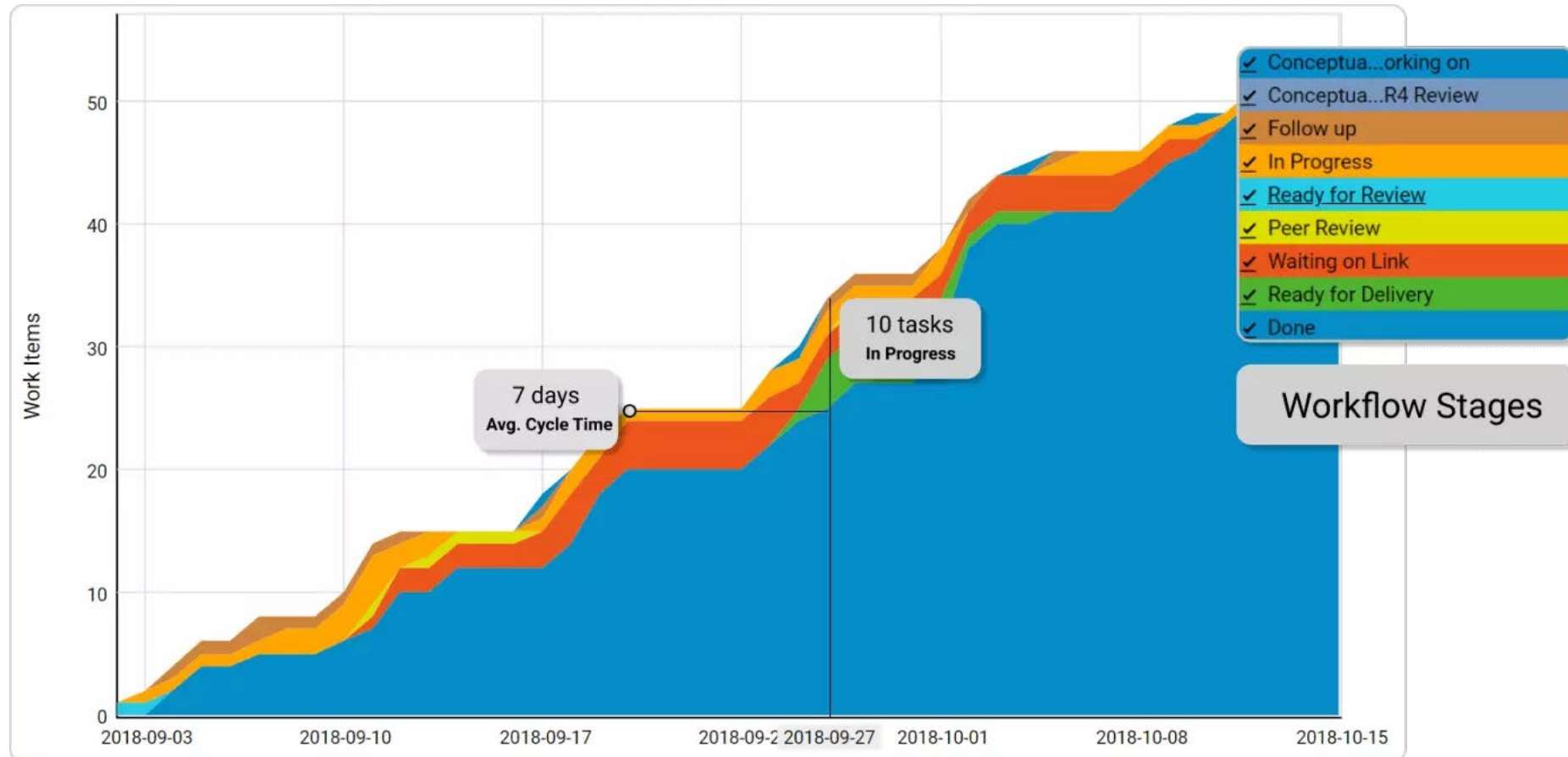
Effectiveness of the Development Process

- **Cumulative Flow Diagram** monitor the progress and changes of user stories, across different statuses
- Ideally, the curve should be smooth
 - A jagged diagram indicates project bottlenecks like inadequate resources
- This metric works best for an overall and highly visual report



Cumulative Flow Diagram

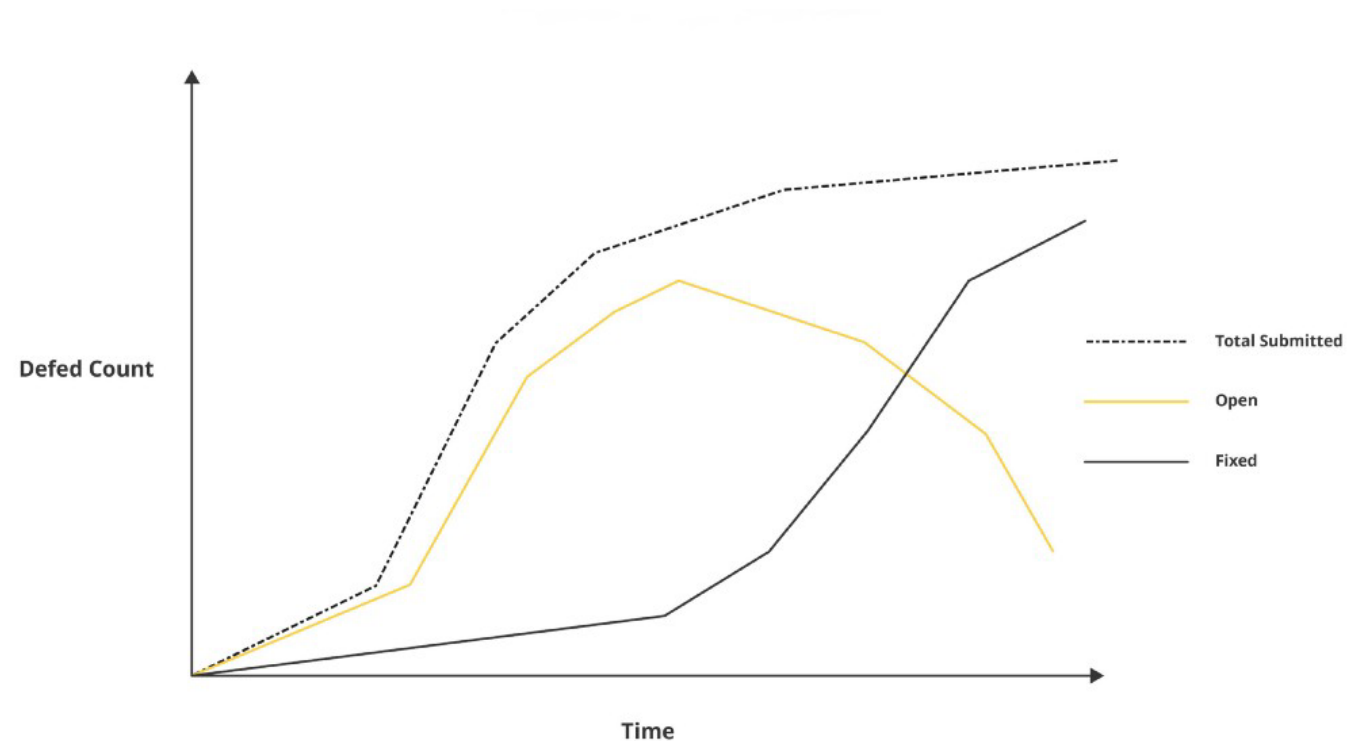
Effectiveness of the Development Process



Defect Trend

Effectiveness of the Development Process

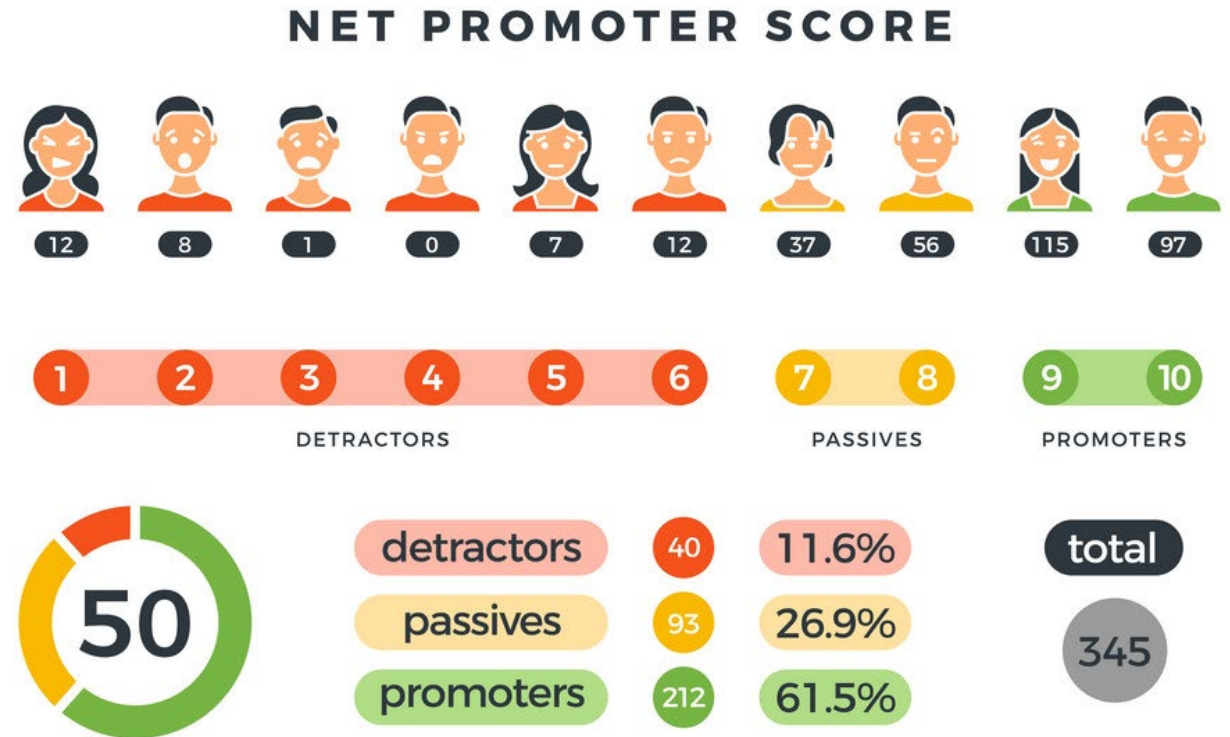
- Using this metric, developers can focus on improving **code quality** as the software release deadline approaches
- Defect here refers to coding bugs in the codebase that could affect the way a software performs, and how customers perceive it



Net Promoter Score

Effectiveness of the Development Process

- One of the principles of agile methodology is to deliver working software that satisfy customers
- And that's what NPS reports on: customer's satisfaction and if they'll recommend the software to others





...

Questions?

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Exercise

Based on the Project Document you delivered in the previous assignment, create a progress report after a first delivery (or milestone).

The report should contain:

- Period covered
- Summary of progress
- Milestone/deliverable completed
- Effort and costs to date and predicted
- Problems encountered*
- Identify potential control actions*

Exercise

Consider that an unexpected event or problem arose during the project. Describe the problem and the impact to the project (using the iron triangle). Also identify and describe potential control actions to solve this problem.

Use the last digit on your day of birth for selecting one of the problems:

1-2. Changes to project scope: new requirements or changes in stakeholder expectations affected the scope

3-4. Technical issues: software bugs or hardware failures may have arisen that required additional time and resources to resolve

5-6. Staffing issues: a team member leaved the project unexpectedly or did not meeting performance expectations

7-8. Resource constraints: a lack of budget or unexpected delays in obtaining necessary resources

9-10. Stakeholder issues: disagreements over project goals or unforeseen challenges with communication

Bibliography

- Cadle, James, and Donald Yeates, eds. **Project management for information systems**. Pearson education, 2004.
- Jira Software Support. **Track and Analyze your team's work with reports**. Available at <https://support.atlassian.com/jira-software-cloud/docs/track-and-analyze-your-teams-work-with-reports/>